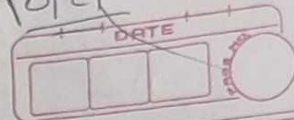


10th Feb
2026

Experiment no-3

Expt

10/2/26



Determination of Hardness of water:

given: volume of hard water = 10 mL
molar molarity of EDTA = 0.01 M

Part A:

Calculation

1 mole EDTA = 1 mole Ca^{2+} or 1 mole Mg^{2+}
1 mole CaCO_3

1 mL, 1 M EDTA = 100 mg CaCO_3 equivalent

Total Hardness = $\frac{\text{volume of EDTA required} \times \text{molarity of EDTA} \times 10^5}{\text{volume of hard water}}$

$$= \frac{10 \text{ mL} \times 0.01 \times 10^5}{10} = \frac{10 \times 0.01 \times 10^5}{10}$$

$$= \underline{100} \text{ ppm or mg/Lit}$$

Part B: Permanent hardness of given water sample: EDTA from burette, $V_2 =$ mL

$$\text{Permanent Hardness} = \frac{V_2 \times 0.01 \times 10^5}{10} = \frac{5 \times 0.01 \times 10^5}{10}$$

$$= \underline{500} \text{ ppm or mg/Lit}$$

Part C: Temporary hardness of given water sample:

Temporary hardness = Total hardness - Permanent Hardness

$$= 1000 - 500 = \underline{500} \text{ ppm or mg/Lit}$$